INDIAN INSTITUTE OF TECHNOLOGY ROORKEE

Distributed Systems

Introduction

by Manoj Misra, IIT Roorkee

Course Content

Introduction to Distributed Systems, Basic Algorithms in Message Passing System, Leader Election in Rings, Distributed Minimum Spanning Tree, Models of Distributed Computation, Causality & Logical Time, Global State and Snapshot Recording Algorithms, Distributed Mutual Exclusion Algorithms, Distributed Shared Memory, Consensus and Agreement Algorithms, Impossibility of distributed consensus with one faulty process, Fault-tolerance, PAXOS and Friends, Distributed Data Analytics: MapReduce, Distributed Machine Learning, Blockchain Consensus: Introduction to Blockchain Consensus, blockchain, Bitcoin blockchain consensus, The CAP Theorem, Advanced Topics: Edge Computing

Books

1. Kskhemkalyani, A. and Singhal, M. Distributed Computing: Principles, Algorithms and Systems. Cambridge University Press
2. Attiya, Hagit, and Jennifer Welch. Distributed computing: fundamentals, simulations, and advanced topics. *Vol. 19. John Wiley & Sons, 2004*.
3. Lynch, Nancy A. Distributed algorithms. *Elsevier, 1996*.
4. Coulouris G, Dollimore J, Kindberg T and Blair G, Distributed Systems: Concepts and Design, Pearson Education
5. Tanenbaum A S and Steen M V, Distributed Systems: Principles and Paradigms, Pearson Education

Background Needed

- Data Structures and Algorithms
- Computer Networks
- Operating Systems
- Database management systems
- Unix
- C / C++
- Socket API
- RPC

What is a distributed system?

- A distributed system is a 1 collection of independent computers that appear to the users of the system as a single computer. 2 [Andrew Tanenbaum]

  Ameoba

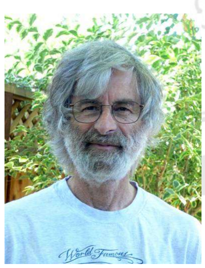
- 3 A distributed system is a collection of processors that do not share memory or clock. Instead each processor has its own local memory and the processors communicate with one another through communication lines. 4 [Galvin et al.]

- System of networked computers that
  - communicate and coordinate their actions only by passing messages

  [Coulouris et al.]

What is a distributed system?

"A distributed system is one in which the failure of a computer you didn't even know existed can render your own computer unusable"



Turing

-Leslie Lamport

What is a distributed system?

- ✓ ~~Concurrency~~ — Adv.
- ✓ No global clock — issue
- ✓ Independent failure

[Coulouris et al.]

- A collection of mostly autonomous processors communicating over a communication network and having the following features

- ✓ No common physical clock
- ✓ No shared memory
- ✓ Geographical separation
- ✓ Autonomy and heterogeneity

[Singhal and Kskhemkalyani]

What is a distributed system?

- A practical "distributed system" will probably have both
  - ✓ Processes on different computers that communicate by messages
  - ✓ Processes/threads on a computer that communicate by messages or shared memory — pipes
    ipc — message queues

A very broad definition:

- A set of autonomous processes communicating among themselves to perform a task
  - ✓ Autonomous: able to act independently
  - ✓ Communication: shared memory or message passing

Distributed systems

A typical distributed system

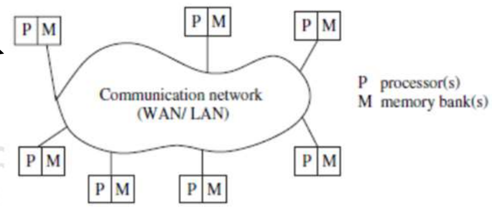


Fig 1

Relationships of the software components that run on each of the computers and use the local operating system and network protocol stack for functioning.

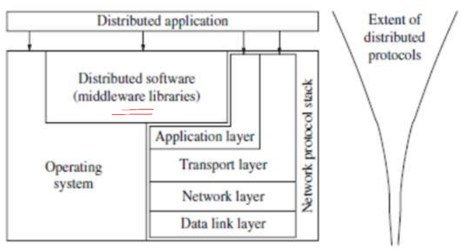


Fig 2

Middleware:

CORBA,
RPC
DCOM
Java, and RMI
MPI

IIT ROORKEE 9

Flynn's taxonomy

Feng

- Single Instruction Single Data (SISD)
- Single Instruction Multiple Data (SIMD)
- Multiple Instruction Single Data (MISD)
- Multiple Instruction Multiple Data (MIMD)

IIT ROORKEE 10

Why distributed systems?

- **Nature of the application**
  - ✓ *Multiplayer games, P2P file sharing, banking, or reaching consensus among parties that are geographically distant*
- **Availability despite unreliable components / Enhanced reliability**
- **Scale up capacity**
  - – Processing
  - – Storage

  Individual computers have limited resources compared to scale of current day problems & application domains

Why distributed systems?

- Resource sharing: peripherals, data sets, libraries
  - ▪ Access to geographically remote data and resources
- Conquer geographic separation
  - ✓ *A web request in India is faster served by a server in India than by a server in US.*
- Increased performance/cost ratio
- Modularity and incremental expandability

Question

Q. Which of the following is not true for a Distributed System:

(a) No global physical clock T

(b) Communication (mostly) through message passing T

(c) No resource sharing F

(d) Concurrent execution T